



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
DEPARTMENT OF
University
Hosur, 100 Km from U.C.C. Area, 100 Km
Avadi, Chennai

SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ASSIGNMENT

Name of the Student	C. Jaya Prakash
VTU No	25513
Semester / Year /Slot	WS-2526/2026/TP1
Department	CSE
Subject Code / Name	10212CS214/Data Visualization
Task No	5
Date of Announcement	08-04-26
Date of Submission	15-04-26

Assignment Mark Allocation

Assignment Mark Allocation	Total Mark	Mark Achieved
Concept Understanding (A)	1	1
Accuracy & Correctness (B)	1	1
Presentation Clarity & Communication (C)	1	1
Relevance & Completion (D)	1	1
Timeliness (E)	1	1
Total (A + B + C + D + E)	5	5

General Comments:

Faculty Signature:

Faculty Name: Mrs. Suganya V.



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(Established by UGC Act, 1996)
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COs	Course Outcomes	K – Level
CO1	Explain the visualization process and identify the type of data	K2
CO2	Apply various visualization techniques for Exploratory Data Analysis	K3
CO3	Explore visualization techniques Trees, Graphs, Networks, Text and documents	K3
CO4	Utilize various visualization techniques for spatial and Geospatial data	K3
CO5	Make use of different visualization tools for various applications	K3
Knowledge Level (Based on revised Bloom's Taxonomy) K1-Remember K2-Understand K3-Apply K4-Analyze K5-Evaluate K6-Create		

Question Set:

TASK-			
S.No	List of Questions	CO	K Level
1	A hospital aims to analyze patient data to identify common diseases and improve treatment efficiency. Create a dataset, perform data cleaning, generate summary statistics, derive new features (e.g., risk score), and detect anomalies in patient records.	CO1	K2
2	A hospital aims to analyze patient data to identify disease patterns and admission trends. Use Visualization techniques such as bar charts, pie charts, scatterplots, and density plots to represent relationships between variables.	CO2	K3
3	A hospital aims to analyze patient data across diseases, age groups, and treatment periods. Perform Multivariate analysis using parallel coordinates, scatterplot matrices, line graphs, and stacked bar charts.	CO3	K3
4	A hospital aims to visualize patient distribution across different locations to improve healthcare planning. Use Geospatial visualization techniques such as scatter maps and spatial heatmaps using location data.	CO4	K3
5	A hospital aims to analyze admission trends over time to predict future demand. Use Forecasting techniques such as line plots, trend lines, and time-series analysis to study seasonal variations and forecast future admissions.	CO5	K3

Set Creation, Data Cleaning, Summary Statistics, Feature Engineering, Anomaly Detection.

In healthcare systems, Patient data analysis is a crucial role in identifying disease patterns and improving treatment efficiency. The first step is to create a structured dataset.

DataSet Creation:

A hospital data set may include the following attributes:

- Patient-ID
- Age
- Gender
- Disease
- Blood Pressure
- Sugar Level
- Admission Date
- Location

This dataset represents Patient health records used for analysis.

Data Cleaning:

Raw data is often incomplete and inconsistent. Data cleaning ensures quality.

Steps:

- Handle missing values using mean/median
- Remove duplicate records
- Standardize formats (male/female instead of M/F)
- Convert data types (dates, numbers)
- Detect and remove outliers

→ Clean data improves accuracy of analysis

Feature Engineering

New features are created to improve analysis.

Example:

$$\rightarrow \text{Risk Score} = (\text{BP} + \text{Sugar} + \text{Age}) / 3$$

helps Classify Patients into:

- Low risk
- medium risk
- High risk

Raw data



Data cleaning



Summary Statistics



features engineering



Anomaly detection



final Dataset

②

Visualization Techniques (Bar, Pie, Scatter, Density plots)

- A) visualization transforms complex data into graphical form making it easy to understand patterns.

Bar Chart

Used to Compare categories

Example:

• Number of patients per disease

→ Helps identify most common disease

Pie chart

Shows Percentage distribution

Example:

• 40% diabetes

• 30% hypertension

Scatter plot

Shows relationships between two variables

Example:

- Age vs Sugar level
- BP vs Age

→ Helps identify Correlation

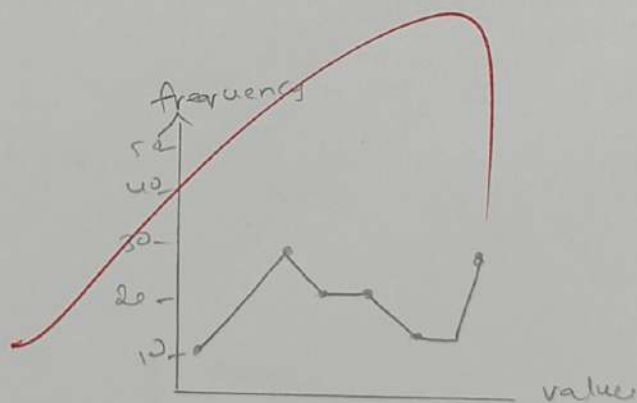
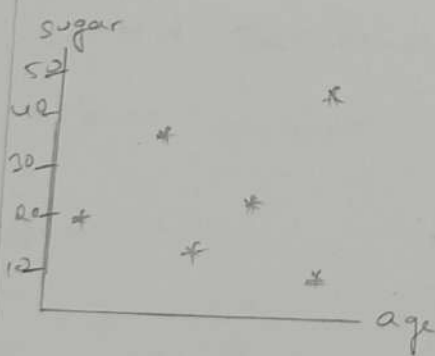
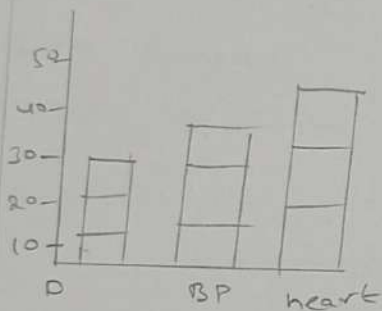
Density plot

Shows distribution of data

Example:

- Distribution of sugar level

→ Helps understand spread and clusters



Multivariate Analysis

① multivariate analysis studies multiple variables together to understand complex relationships

Parallel Coordinates plot

each line = one patient
each axis = one feature

Scatterplot matrix

Shows pairwise relationships between variables

Example:

- age vs BP
- BP vs Sugar

Line Graph

Used to show trends over time

Example:

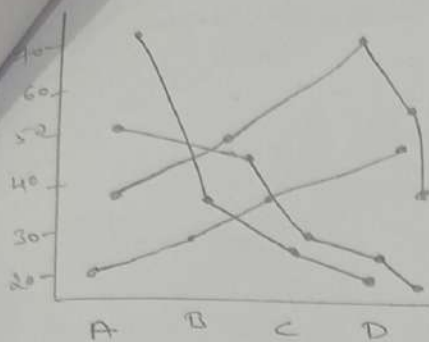
- Daily admissions

Stacked Bar Chart

Shows multiple categories

Example:

- Disease distribution by gender

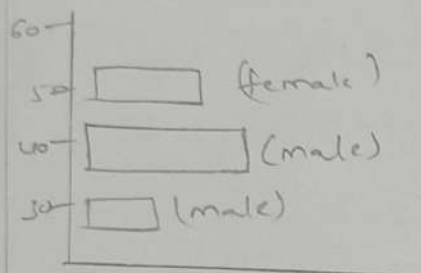


Parallel Coordinates

[Age vs BP] [Age vs sugar]

[BP vs sugar] [sugar vs Age]

Scatter matrix



Stacked Bar

k) Geospatial Visualization

- Ⓐ Geospatial visualization uses location data to analyze patient distribution.

Scatter map

Shows Patient Locations

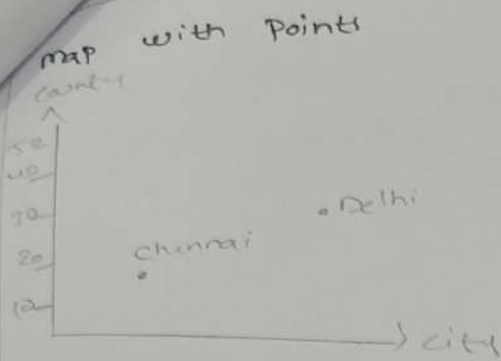
Example:

- Chennai, Delhi, Mumbai

Heatmap

Shows density of patients

→ Dark color = more patient



Heat map

[Light] $\rightarrow$ Low case

[Dark] $\rightarrow$ high case

5) Forecasting Techniques

forecasting Predicts future patient admissions using historical data

Time - Series Analysis

Data collected over time:

- Daily admissions
- monthly cases

Line plot

Shows trend over time

Trend Line

Shows increasing or decreasing pattern

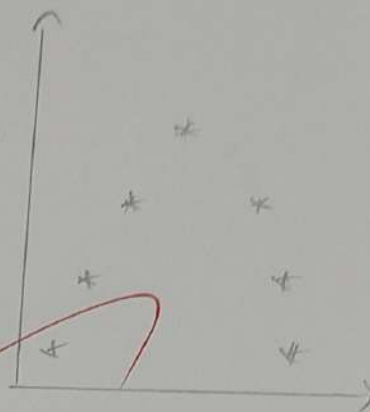
methods:

- moving Average
- ARIMA
- Exponential Smoothing

Seasonal Analysis

- flu increased in winter
- Dengue increases in rainy season.

Patients



86
16/4/26